

Allan Quadros

Department of Management, Coggin College of Business, University of North Florida, Jacksonville, FL

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EDUCATION

Ph.D. in Statistics 2021–2025
Kansas State University

B.Sc. in Statistics 2014–2018
University of Brasilia (UnB)
* Recipient of the ‘Outstanding Student Award’ (highest GPA: 4.96/5.00)

M.Sc. in Economic Development 2009–2012
State University of Campinas (Unicamp)

Licenciatura in Geography 2003–2008
State University of Campinas (Unicamp)

ACADEMIC APPOINTMENTS

Instructor of Operations Management and Information Systems Aug 2025 – present
Department of Management, Coggin College of Business, University of North Florida

Risk Modeling Consultant, NBAF/USDA Project May 2024 – Jul 2025
College of Business, Kansas State University
Developed the statistical methodology for predicting time-to-failure without prior failure data for the National Bio and Agro-Defense Facility, enabling a risk-based maintenance policy supporting U.S. Congress budget requests. Funded by two USDA grants totaling \$271,385.

Graduate Teaching Assistant Apr 2021 – May 2025
Department of Statistics, Kansas State University
Instructor of record for undergraduate statistics courses; specialized in computational statistics, Bayesian hierarchical modeling, resampling methods, and survival analysis.

Lecturer in Economics Jul 2010 – Apr 2013
University Center of Jaguariuna (UniFAJ), Brazil

RESEARCH GRANTS

CCB Advancement Grant, University of North Florida Mar 2026
\$12,000 in support of “Detecting False Positives in Cointegration Tests: A Bayesian Prior-Posterior Conflict Approach”.

CCB Societal Impact Faculty Development Grant, UNF Jan 2026
\$4,000 in support of “Improving Chronic Disease Management: Spatiotemporal Patterns of Care Delivery Before and During the Pandemic”.

NBAF Life Cycle of Equipment, USDA (Lead Developer) 2023–2024
Two grants totaling \$271,385 supporting the development of a Bayesian methodology for equipment failure prediction at the National Bio and Agro-Defense Facility.

PUBLICATIONS

Peer-Reviewed Articles

- [1] **Quadros, A.**, Higgins, M., & Silverstein, B. (2026). A Bayesian approach to generating distribution-based signals in pairs trading. *Quantitative Finance*, 1–21. doi.org/10.1080/14697688.2026.2644357
- [2] Souza, P., **Quadros, A.**, Dogan, H., Li, Y., Shi, Y. C., & Karkle, E. (2026). Exploring bread quality through the use of commercial bacterial and fungal xylanases: Effects on dough rheology, loaf volume, and arabinoxylan structure. *Journal of Food Science*, 91(2), e70940. doi.org/10.1111/1750-3841.70940
- [3] Cançado, A., Oliveira, G., **Quadros, A.**, & Duczmal, L. (2025). A tree-spatial scan statistic. *Environmental and Ecological Statistics*, 32, 953–978. doi.org/10.1007/s10651-025-00670-w
- [4] **Quadros, A.** (2024). mRpostman: An IMAP client for R. *Journal of Open Research Software*, 12(1), 4. doi.org/10.5334/jors.480
- [5] von Borries, G., & **Quadros, A.** (2022). ROC App: An application to understand ROC curves. *Brazilian Journal of Biometrics*, 40(2). doi.org/10.28951/bjb.v40i2.566
- [6] Beaklini, P., **Quadros, A.**, Avellar, M., Dantas, M., & Cançado, A. (2020). AGN dichotomy beyond radio loudness: A Gaussian mixture model analysis. *Monthly Notices of the Royal Astronomical Society*, 497(2), 1463–1474. doi.org/10.1093/mnras/staa2072

Manuscripts Under Review

- [1] **Quadros, A.**, Higgins, M., & Silverstein, B. (2026). Distribution-based algorithms for pairs trading: A block bootstrap approach. Submitted to *Quantitative Finance*.

Manuscripts in Preparation

- [1] **Quadros, A.** Detecting false positives in cointegration tests: A Bayesian prior-posterior conflict approach. Funded by the UNF/CCB Advancement Grant.
- [2] Shang, D., & **Quadros, A.** Improving chronic disease management: Spatiotemporal patterns of care delivery before and during the pandemic. Funded by the UNF/CCB Societal Impact Faculty Development Grant.
- [3] **Quadros, A.**, Higgins, M., & Silverstein, B. Bayesian estimation of equipment failure risk in the absence of historical failure data. Targeting *IEEE Transactions on Reliability*.

Books and Textbooks

- [1] **Quadros, A.** (2025). Introduction to Data Analysis with R and the Tidyverse. Online textbook used in Data Analytics and Applications of Business Analytics courses at UNF. allanvc.github.io/book_IDART_en
- [2] **Quadros, A.** (2020). Introdução à Análise de Dados em R utilizando Tidyverse. allanvc.github.io/book_IADR-T
- [3] **Quadros, A.** Who is Afraid of Math? (In preparation).

CONFERENCE PRESENTATIONS

- [1] **Quadros, A.**, Higgins, M., & Silverstein, B. (2026, June). Distribution-based algorithms for pairs trading: A block bootstrap approach. Forthcoming, Quantitative Finance and Risk Analysis Symposium 2026 (QFRA 2026), Zakynthos, Greece.
- [2] **Quadros, A.**, Shang, D., & Williams, C. (2026, June). Improving chronic disease management: Spatiotemporal patterns of care delivery before and during the pandemic. Forthcoming, 34th European Conference on Information Systems (ECIS 2026), Milan, Italy.

[3] **Quadros, A.**, Higgins, M., & Silverstein, B. (2025, July). A Bayesian approach to generating distribution-based signals in pairs trading. 2025 SIAM Conference on Financial Mathematics and Engineering (SIAM FM25), Miami, FL.

TEACHING

University of North Florida (Coggin College of Business)

ISM6404 – Applications of Business Analytics (MSBA)	Fall 2025, Spring 2026
CAP6768 – Data Analytics (MSBA)	Fall 2025, Spring 2026
MAN4583 – Project Management (Undergraduate)	Fall 2025, Spring 2026
MAN4550 – Introduction to Management Science (Undergraduate)	Summer 2026
ISM4210 – Information and Data Management (Undergraduate)	Spring 2026

Kansas State University (Department of Statistics)

STAT341 – Biometrics II (Instructor)	Spring 2024, 2025
STAT340 – Biometrics I (Instructor)	Spring 2023, Fall 2024
STAT325 – Introduction to Statistics (Instructor)	2021–2022
STAT703 – Introduction to Statistical Methods for the Sciences (TA)	Summer 2023, 2024
STAT351 – Business and Economic Statistics II (TA)	Fall 2023

Invited Lecturer (Brazil)

Inferential Statistics with Python – MBA, ENAP	Spring 2025
Predictive Analysis – Machine Learning – MBA, FGV Management	Fall 2021 – Fall 2024
Exploratory Data Analysis – MBA, FGV Management	Fall 2021 – Fall 2024
Advanced Predictive Analysis – Deep Learning – MBA, FGV Management	Spring 2022
Data Analysis with R (Introductory & Intermediate) – FNDE	2018–2019

SOFTWARE DEVELOPMENT

Open-Source R Packages

treeSS (2026) – Tree-Spatial Scan Statistic for Cluster Detection. Available on CRAN. CRAN | GitHub

mRpostman (2019) – An IMAP client for R. Selected as one of the top 40 new CRAN packages of August 2019 by RStudio/Posit's R Views Blog. CRAN | GitHub | Website

emstreeR (2018) – Tools for Fast Computation and Visualization of Euclidean Minimum Spanning Trees. Available on CRAN. CRAN | GitHub

onlineretail (2020) – Online retail dataset. Available on CRAN. CRAN | GitHub

Proprietary Software

NBAF Equipment Failure Prediction (2025) – Software and statistical methodology for estimating time-to-failure of large industrial equipment. Kansas State University / NBAF / USDA. Language: R.

Malha Fina (2020) – Financial accountability and fraud detection software. Ministry of Education / FNDE, Brazil. Language: Python.

proPNLD (2019) – Forecasting solution for student enrollment by grade and school across the Brazilian public education system. Ministry of Education / FNDE, Brazil. Language: R.

FisFa – Fiscaliza Fatura (2016) – Financial accountability and contract management software credited with over US\$ 50 million in savings to the Brazilian federal government (2015–2018). Ministry of Education / FNDE, Brazil. Language: Python. Brazilian Patent BR 51 2016 000212 3.

SERVICE

Co-Organizer, Management Research Seminar Series, Department of Management, University of North Florida (with Pingying Zhang). Aug 2025 – present

OTHER RELEVANT PROFESSIONAL EXPERIENCE

Quantitative Researcher & Developer Jun 2022 – Jun 2024
BX Analytics (hedge fund), Campinas, Brazil
Designed and deployed two statistical arbitrage methodologies and accompanying software solutions for trading strategies and portfolio management.

Lead Data Scientist Apr 2020 – Dec 2020
National Fund for Educational Development (FNDE), Ministry of Education, Brazil
Led the data science team in implementing machine and statistical learning solutions; advocated successfully for organization-wide adoption of data-driven decision making.

Data Scientist May 2013 – Apr 2020
National Fund for Educational Development (FNDE), Ministry of Education, Brazil
Authored FisFa (over US\$ 50M in federal savings, 2015–2018), proPNLD (national textbook procurement forecasting), and Malha Fina (anomaly-detection-based audit system).

First Lieutenant Feb 2003 – Jun 2009
Brazilian Army, Infantry Branch
Led a platoon of 30+ personnel; instructor of military courses; public relations officer.